

Qualification Pack



Automobile Technology

QP Code: ASC/Q1439

Version: 1.0

NSQF Level: 4.5

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Qualification Pack

Contents

ASC/Q1439: Automobile Technology	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	3
ASC/N9828: Work organization, management and safety (Automobile Technology)	5
ASC/N9829: Communication skills (Automobile Technology)	8
ASC/N1476: Service	11
ASC/N1477: Electrical and mechanical systems inspection and diagnosis	14
ASC/N1478: Repair and overhaul	17
Assessment Guidelines and Weightage	19
<i>Assessment Guidelines</i>	19
<i>Assessment Weightage</i>	20
Acronyms	21
Glossary	22

Qualification Pack

ASC/Q1439: Automobile Technology

Brief Job Description

The Light Vehicle Automotive Technician will work in the service, diagnosis and repair sector of light vehicles in either independent workshops or the vehicle manufacturers affiliated workshops. The Light Vehicle Automotive Technician will be able to service, diagnose, and repair a range of light vehicles using service and repair information, and a wide range of diagnostic and service tools.

Personal Attributes

The person should be patient, organised, team-oriented and have the ability to work for long hours in adverse conditions. They should be keen observers and have an eye for detail and quality.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [ASC/N9828: Work organization, management and safety \(Automobile Technology\)](#)
2. [ASC/N9829: Communication skills \(Automobile Technology\)](#)
3. [ASC/N1476: Service](#)
4. [ASC/N1477: Electrical and mechanical systems inspection and diagnosis](#)
5. [ASC/N1478: Repair and overhaul](#)

Qualification Pack (QP) Parameters

Sector	Automotive
Sub-Sector	Automotive Vehicle Service
Occupation	Technical Service & Repair
Country	India
NSQF Level	4.5
Credits	17
Aligned to NCO/ISCO/ISIC Code	NCO-2015/3115.0602

Qualification Pack

Minimum Educational Qualification & Experience	No formal education prescribed
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	25 Years
Last Reviewed On	NA
Next Review Date	08/02/2026
NSQC Approval Date	08/02/2024
Version	1.0
Reference code on NQR	QG-4.5-AU-01832-2024-V1-ASDC
NQR Version	1

Qualification Pack

ASC/N9828: Work organization, management and safety (Automobile Technology)

Description

This NOS unit is about implementing safety, planning work, adopting sustainable practices for optimising the use of resources.

Scope

The scope covers the following :

- Manage work and resources

Elements and Performance Criteria

Manage work and resources

To be competent, the user/individual on the job must be able to:

- PC1.** Prepare and maintain safe, tidy, and efficient workstations
- PC2.** Plan, prepare, and complete each task within the time available
- PC3.** Select, use, clean, store, and test all equipment and materials safely and in accordance with manufacturers' instructions
- PC4.** Apply the health, safety, and environmental standards within the automotive industry
- PC5.** Reinstatement work areas and vehicles to an appropriate condition

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Working safely, including: Procedures, Housekeeping, Hygiene
- KU2.** The purpose, use, care, maintenance, and safety implications of: Equipment, Materials, Fluids
- KU3.** The risks of the automotive service repair industry, including: Causes, Prevention
- KU4.** Time management
- KU5.** The environment, including: Sustainability
- KU6.** Waste management

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Manage work and resources</i>	15	25	-	10
PC1. Prepare and maintain safe, tidy, and efficient workstations	3	5	-	2
PC2. Plan, prepare, and complete each task within the time available	3	5	-	2
PC3. Select, use, clean, store, and test all equipment and materials safely and in accordance with manufacturers' instructions	3	5	-	2
PC4. Apply the health, safety, and environmental standards within the automotive industry	3	5	-	2
PC5. Reinststate work areas and vehicles to an appropriate condition	3	5	-	2
NOS Total	15	25	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N9828
NOS Name	Work organization, management and safety (Automobile Technology)
Sector	Automotive
Sub-Sector	Generic
Occupation	Generic
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2026
NSQC Clearance Date	08/02/2024

Qualification Pack

ASC/N9829: Communication skills (Automobile Technology)

Description

This unit is about communicating with team members, superior and others.

Scope

The scope covers the following :

- Communicate effectively with others

Elements and Performance Criteria

Communicate effectively with others

To be competent, the user/individual on the job must be able to:

- PC1.** Communicate with clarity, effectiveness, and efficiency in the workplace, including: Written, Verbal, Non-verbal, Electronic means, Use of technology, Using standard formats
- PC2.** Read, interpret, and extract technical data and instructions
- PC3.** Use digital or manual systems to access documents
- PC4.** Complete reports and record keeping
- PC5.** Respond to customers' needs face to face and indirectly

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Communication, including: Electronic means Written, Verbal, Non-verbal, Use of technology
- KU2.** Technical language
- KU3.** Technical drawings
- KU4.** Schematics
- KU5.** Wiring diagrams
- KU6.** Record keeping and reporting
- KU7.** Customer service

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Communicate effectively with others</i>	15	25	-	10
PC1. Communicate with clarity, effectiveness, and efficiency in the workplace, including: Written, Verbal, Non-verbal, Electronic means, Use of technology, Using standard formats	3	5	-	2
PC2. Read, interpret, and extract technical data and instructions	3	5	-	2
PC3. Use digital or manual systems to access documents	3	5	-	2
PC4. Complete reports and record keeping	3	5	-	2
PC5. Respond to customers' needs face to face and indirectly	3	5	-	2
NOS Total	15	25	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N9829
NOS Name	Communication skills (Automobile Technology)
Sector	Automotive
Sub-Sector	Generic
Occupation	Generic
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2026
NSQC Clearance Date	08/02/2024

Qualification Pack

ASC/N1476: Service

Description

This NOS is about performing routine servicing of the vehicle.

Scope

The scope covers the following :

- Routine servicing of vehicle

Elements and Performance Criteria

Routine servicing of vehicle

To be competent, the user/individual on the job must be able to:

- PC1.** Perform vehicle servicing using the standards and procedures developed by the automobile industry and vehicle manufacturers as required on the above listed systems, including:
- PC2.** Inspection
- PC3.** Adjustment
- PC4.** Renewing or replacement
- PC5.** Tensioning
- PC6.** Lubrication
- PC7.** Use relevant tools and equipment

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Scheduled maintenance vehicle servicing
- KU2.** The use and operation of relevant tools and equipment
- KU3.** The servicing requirements and procedures for vehicle systems, including: Engine mechanical, Driveline and final drives, Steering and suspension, Brakes, Wheels and tyres, Battery and electrical, Fuel, Cooling

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Routine servicing of vehicle</i>	15	25	-	10
PC1. Perform vehicle servicing using the standards and procedures developed by the automobile industry and vehicle manufacturers as required on the above listed systems, including:	2	3	-	1
PC2. Inspection	2	3	-	1
PC3. Adjustment	2	4	-	2
PC4. Renewing or replacement	2	4	-	2
PC5. Tensioning	2	3	-	1
PC6. Lubrication	3	5	-	2
PC7. Use relevant tools and equipment	2	3	-	1
NOS Total	15	25	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N1476
NOS Name	Service
Sector	Automotive
Sub-Sector	Automotive Vehicle Service
Occupation	Technical Service & Repair
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2026
NSQC Clearance Date	08/02/2024

Qualification Pack

ASC/N1477: Electrical and mechanical systems inspection and diagnosis

Description

This NOS is about performing inspection and diagnosis of vehicle electrical and mechanical systems,

Scope

The scope covers the following :

- Electrical and mechanical systems inspection and diagnosis

Elements and Performance Criteria

Electrical and mechanical systems inspection and diagnosis

To be competent, the user/individual on the job must be able to:

- PC1.** Apply the problem-solving techniques used in manufacturers' diagnostic procedures to identify the condition, cause, and correction of light vehicle faults in the above listed systems, including: Verify the condition, including: Visual inspection and basic tests
- PC2.** Diagnose the cause, including: Using mechanical and electrical test equipment, retrieving scan tool data, Check and interpret results, recommend corrective action

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The principles and applications of specialist diagnostic procedures
- KU2.** The use and operation of relevant tools, diagnostic tools and equipment
- KU3.** The principles and applications of condition, cause and correction in fault diagnosis
- KU4.** The construction and operation of the following systems: Spark ignition engine management, Compression ignition engine management, Engine mechanical, Cooling, Hybrid/electric vehicle, Forced induction, Emission control, Exhaust, Electrical/electronics, Braking and stability control, Suspension and steering, Advanced driver assistance systems (ADAS), Drive train, Heating ventilation air conditioning (HVAC), Air bag / supplementary restraint systems (SRS), Infotainment and Communication systems

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Electrical and mechanical systems inspection and diagnosis</i>	6	10	-	4
PC1. Apply the problem-solving techniques used in manufacturers' diagnostic procedures to identify the condition, cause, and correction of light vehicle faults in the above listed systems, including: Verify the condition, including: Visual inspection and basic tests	3	5	-	2
PC2. Diagnose the cause, including: Using mechanical and electrical test equipment, retrieving scan tool data, Check and interpret results, recommend corrective action	3	5	-	2
NOS Total	6	10	-	4

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N1477
NOS Name	Electrical and mechanical systems inspection and diagnosis
Sector	Automotive
Sub-Sector	Automotive Vehicle Service
Occupation	Technical Service & Repair
NSQF Level	4.5
Credits	7
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2026
NSQC Clearance Date	08/02/2024

Qualification Pack

ASC/N1478: Repair and overhaul

Description

This NOS unit is about repairing and overhauling of the vehicle.

Scope

The scope covers the following :

- Repairing and overhauling

Elements and Performance Criteria

Repairing and overhauling

To be competent, the user/individual on the job must be able to:

- PC1.** Perform vehicle repairs using the standards and procedures developed by the automobile industry and vehicle manufacturers on the above listed systems, including:
- PC2.** Removal and replacement of components
- PC3.** Disassembly and reassembly of components
- PC4.** Replacement of parts
- PC5.** Overhaul of components
- PC6.** Measurement and adjustment of components
- PC7.** Testing of completed repair

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The principles and applications of repair and overhaul procedures
- KU2.** The use and operation of relevant tools, special tools and equipment
- KU3.** The repair methods and procedures of the following systems: Spark ignition engine management, Compression ignition engine management, Engine mechanical, Cooling, Hybrid/electric vehicle, Forced induction, Emission control, Exhaust, Electrical/electronics, Braking and stability control, Suspension and steering, Advanced driver assistance systems (ADAS), Drive train, Heating ventilation air conditioning (HVAC), Air bag / supplementary restraint systems (SRS), Infotainment and Communication systems

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Repairing and overhauling</i>	15	25	-	10
PC1. Perform vehicle repairs using the standards and procedures developed by the automobile industry and vehicle manufacturers on the above listed systems, including:	2	3	-	1
PC2. Removal and replacement of components	2	3	-	1
PC3. Disassembly and reassembly of components	2	4	-	2
PC4. Replacement of parts	2	4	-	2
PC5. Overhaul of components	2	3	-	1
PC6. Measurement and adjustment of components	3	5	-	2
PC7. Testing of completed repair	2	3	-	1
NOS Total	15	25	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N1478
NOS Name	Repair and overhaul
Sector	Automotive
Sub-Sector	Automotive Vehicle Service
Occupation	Technical Service & Repair
NSQF Level	4.5
Credits	4
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2026
NSQC Clearance Date	08/02/2024

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training centre (as per assessment criteria below).
4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training centre based on these criteria.
5. In case of successfully passing only certain number of NOSs, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack.
6. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

Qualification Pack

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ASC/N9828.Work organization, management and safety (Automobile Technology)	15	25	0	10	50	5
ASC/N9829.Communication skills (Automobile Technology)	15	25	0	10	50	10
ASC/N1476.Service	15	25	0	10	50	20
ASC/N1477.Electrical and mechanical systems inspection and diagnosis	6	10	0	4	20	45
ASC/N1478.Repair and overhaul	15	25	0	10	50	20
Total	66	110	-	44	220	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.